```
    float r;

public:
  void init(float x)  /* Inline function */
  {
    r = x;
  }
  float area();
};

float Circle::area()
{
  return 3.14*r*r;
}

int main(int argc, char **argv)
{
  float radius;

  Circle circle;
  cout << "Enter the radius of the circle:: ";
  cin >> radius;
  circle.init(radius);
  cout << "Area of the Circle:: "<<circle.area()<<"\n";

  return 0;
}
```

Save the file in the *codes* sub-directory as *class-demo.cxx*. Compile and execute it:

```
$ g++ -o class-demo class-demo.cxx
$ ./class-demo
Enter the radius of the circle:: 4
Area of the Circle:: 50.24
```

Assuming that you have been able to compile these programs successfully, I would now recommend you go ahead and write, compile and test some of your C/C++ assignments and problems using *gcc* and *g++*. If you face any issues, you are most welcome to e-mail me personally.

Java programming on Linux

Java is perhaps the next most widely taught language in Indian schools and colleges after C/C++. The best part of Java programming on Linux is that you use the same tools that you would use on Windows—yes, the Sun Java Development Kit. To install the JDK on Linux, download the installer for Linux from http://java.sun.com/javase/downloads/widget/jdk6.jsp. Choose the *.bin* file, and not the **rpm.bin* file, unless you know what you are doing. (The *.bin* file is the equivalent of *.exe* on Windows). Once the download is complete, in your terminal, *cd* to the directory where the file has been downloaded, and use the following commands:

```
$ chmod +x jdk-6u18-linux-i586.bin
$ ./jdk-6u18-linux-i586.bin
```

The file names above might differ depending on the JDK version that you have downloaded.

The first line makes the installer executable, and the second line executes it. The installer should start now, and you should see the 'Sun Microsystems, Inc. Binary Code License Agreement'. Accept the licence, and the extraction of the JDK should start. Once the installer has exited, you should see a new sub-directory named 'jdk1.6.0_18' inside the current directory. If you are familiar with Java programming on Windows, this should be easily recognisable. Inside this directory is the *bin* sub-directory, which has the Java compiler (*javac*), Java interpreter (*java*), and others. With this, we are all set; let's write our first Java program on Linux. Fire up *gedit* and write the following Java code, which shows the usage of an array of integers:

```
import java.util.Random;

class ArrayDemo {
    public static void main(String[] args) {
      int[] arr = new int[10];

      for(int i=0;i<10;i++)
          arr[i] = (new Random()).nextInt();

      for(int i=0;i<10;i++)
          System.out.println("Element at index " + i + "is::" + arr[i]);
    }
}
```

Save the code to a file 'ArrayDemo.java', then compile and run it as follows:

```
$ /home/amit/jdk1.6.0_18/bin/javac ArrayDemo.java
$ /home/amit/jdk1.6.0_18/bin/java ArrayDemo
Element at index 0is:: 480763582
Element at index 1is:: -1644219394
Element at index 2is:: -67518401
Element at index 3is:: 619258385
Element at index 4is:: 810878662
Element at index 5is:: 1055578962
Element at index 6is:: 1754667714
Element at index 7is:: 503295725
Element at index 8is:: 1129666934
Element at index 9is:: 1084281888
```

Note the first two commands, where I have given the full path to the location of the *javac* and *java* executables. Depending on where you have extracted the JDK, your path will vary. This is how you can compile, run, test and debug your Java programs.

OpenJDK

An article about Java programming in an open source magazine would be incomplete without talking about